

Introduction to RAG Systems

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with large language models. When a user asks a question, the system first retrieves relevant documents from a knowledge base, then feeds them into an LLM to generate an accurate, grounded response. This approach significantly reduces hallucination and keeps answers up-to-date without requiring model fine-tuning.

RAG systems typically consist of three main components: a document ingestion pipeline that parses and chunks documents, a vector database that stores embeddings, and a retrieval-and-generation loop that finds relevant chunks and synthesizes answers.

Key Benefits

- Reduced hallucination through grounded generation
- Always up-to-date knowledge without retraining
- Transparent attribution to source documents
- Cost-effective compared to fine-tuning large models

Conclusion

RAG represents a practical middle ground between pure retrieval systems and pure generative models. By grounding generation in real documents, it achieves both accuracy and flexibility. Future work will focus on improving retrieval quality and reducing latency.